INDEPENDENCE OF THE MINIMAL-GOOD-LATTICE SPLITTING

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Abstract. We prove that the splitting of the universal metaplectic extension obtained from a minimal good lattice is independent of the choice of that lattice. The proof separates the conjugacy supplied by lattice-model naturality from the equality supplied by the commuting theorem for the two pullback covers.

1. Introduction

Let F be a non-archimedean local field of odd residual characteristic, and let (G, G') be the type-I reductive dual pair under consideration inside the symplectic group of a tensor space W. A generalized lattice model attached to a pair of building points produces a splitting of the universal $\mathbb{C}^\times$-metaplectic extension over the stabilizer of one of those points. We show that, when the point on the G'-side is normalized by a minimal good lattice, the resulting splitting does not depend on the chosen lattice.

The decisive point is that transport by an element of G' first gives a conjugacy identity in the covering group. Passing from conjugacy to equality uses the separate theorem that the pullback covering groups over G and G' commute elementwise. Thus twisted transport alone does not identify the two lattice-model operators.

2. Setup and known inputs

Let

$$(G, G') = (U(V, h), U(V', h')) \subseteq \mathrm{Sp}(W)$$

be the type-I reductive dual pair. We work in the universal extension

$$1 \longrightarrow \mathbb{C}^\times \longrightarrow \widetilde{\mathrm{Sp}}(W) \longrightarrow \mathrm{Sp}(W) \longrightarrow 1.$$

Fix $x \in \mathcal{B}(G)$ and set

$$K_x := \mathrm{Stab}_G(x).$$

The building is refined so that K_x is constant on each open facet. For every $x' \in \mathcal{B}(G')$, the generalized lattice model associated with the tensor lattice

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function $\mathcal{B}_{x,x'}$ supplies a continuous group-homomorphic splitting

$$s_{x,x'}\colon K_x \longrightarrow \widetilde{\mathrm{Sp}}(W).$$

Existence and multiplicativity of these splittings are known inputs.

We use four further known inputs. First, the nonempty set $\mathscr{M}(V')$ of minimal good lattices is one G'-orbit. Every $\Lambda' \in \mathscr{M}(V')$ determines a normalized building point $x'_{\Lambda'}$, and

$$x'_{a\Lambda'} = a\, x'_{\Lambda'} \qquad (a \in G').$$

Second, tensor-lattice equivariance gives

$$\mathcal{B}_{x,ax'} = (1 \otimes a)\mathcal{B}_{x,x'}.$$

Third, transport by $1 \otimes a$ gives an isomorphism

$$T_a\colon \mathcal{H}_{x,x'} \longrightarrow \mathcal{H}_{x,ax'}$$

satisfying, for every $g \in K_x$,

$$T_a M_{x,x'}[g] T_a^{-1} = M_{x,ax'}[g].$$

At cover level, for any lift $\widetilde{a}$ of $1 \otimes a$, this is the conjugacy identity

$$s_{x,ax'}(g) = \widetilde{a}\, s_{x,x'}(g)\widetilde{a}^{-1}.$$

Finally, let $\widetilde{G}$ and $\widetilde{G}'$ denote the pullback covering groups over G and G', respectively. The known commuting-cover theorem states that these two groups commute elementwise inside $\widetilde{\mathrm{Sp}}(W)$. Equivalently, for $g \in G$, $a \in G'$, and arbitrary lifts $\widetilde{g}$ and $\widetilde{a}$,

$$[\widetilde{a}, \widetilde{g}] = 1.$$

This is a theorem about the covering groups and is not inferred from commutativity of G and G' in the base symplectic group.

3. Independence theorem

Theorem 3.1 (Minimal-good-lattice splitting independence)**.** *For $\Lambda' \in \mathscr{M}(V')$, define*

$$s^{\min}_{x,\Lambda'} := s_{x,x'_{\Lambda'}}.$$

For every $\Lambda'_0, \Lambda'_1 \in \mathscr{M}(V')$,

$$s^{\min}_{x,\Lambda'_0} = s^{\min}_{x,\Lambda'_1}$$

as continuous group-homomorphic splittings

$$K_x \longrightarrow \widetilde{\mathrm{Sp}}(W).$$

Hence the common splitting

$$s^{\min}_x\colon K_x \longrightarrow \widetilde{\mathrm{Sp}}(W)$$

is canonically defined.

Proof. Fix $\Lambda_0', \Lambda_1' \in \mathscr{M}(V')$. By the minimal-good-lattice orbit theorem, there is an element $a \in G'$ such that

$$\Lambda_1' = a\Lambda_0'.$$

The normalized building-point equivariance gives

$$x'_{\Lambda_1'} = x'_{a\Lambda_0'} = a\, x'_{\Lambda_0'}.$$

Write $x_0' = x'_{\Lambda_0'}$. Then $x'_{\Lambda_1'} = ax_0'$.

Choose any lift $\widetilde{a}$ of $1 \otimes a$ to $\widetilde{\mathrm{Sp}}(W)$. Tensor-lattice equivariance gives

$$\mathcal{B}_{x,ax_0'} = (1 \otimes a)\mathcal{B}_{x,x_0'}.$$

The naturality input therefore gives, for every $g \in K_x$,

$$T_a M_{x,x_0'}[g] T_a^{-1} = M_{x,ax_0'}[g].$$

At the metaplectic-cover level this is precisely

$$s_{x,ax_0'}(g) = \widetilde{a}\, s_{x,x_0'}(g)\widetilde{a}^{-1}. \tag{1}$$

This is the lattice-model transport step. It does not assert that T_a identifies the two operators inside one fixed Heisenberg model: T_a is twisted by the symplectic action $1 \otimes a$.

For $g \in K_x$, the element $s_{x,x_0'}(g)$ is a lift of g and hence belongs to the pullback cover over G. The element $\widetilde{a}$ belongs to the pullback cover over G'. The known commuting-cover theorem thus gives

$$[\widetilde{a}, s_{x,x_0'}(g)] = 1.$$

Consequently,

$$\widetilde{a}\, s_{x,x_0'}(g)\widetilde{a}^{-1} = s_{x,x_0'}(g). \tag{2}$$

This is a separate cover-level step. It uses the commuting-cover theorem and not merely commutativity of the two groups in the base symplectic group.

Combining (1), (2), and

$$s^{\min}_{x,\Lambda_1'} = s_{x,x'_{\Lambda_1'}} = s_{x,ax_0'},$$

we obtain, for every $g \in K_x$,

$$s^{\min}_{x,\Lambda_1'}(g) = s_{x,ax_0'}(g) = s_{x,x_0'}(g) = s^{\min}_{x,\Lambda_0'}(g).$$

Therefore

$$s^{\min}_{x,\Lambda_0'} = s^{\min}_{x,\Lambda_1'}$$

as maps from K_x to $\widetilde{\mathrm{Sp}}(W)$. Each map $s_{x,x'}$ supplied by the generalized lattice model is already continuous, group-homomorphic, and a splitting. Hence the common map

$$s^{\min}_x := s^{\min}_{x,\Lambda'}$$

has all three properties for every $\Lambda' \in \mathscr{M}(V')$. Its independence of the choice of Λ' follows from the equality just proved, so the splitting is canonical. $\square$

Remark 3.2. The conclusion follows from the stated generalized-lattice-model properties and the known commuting-cover theorem. The latter is a nontrivial input and is deliberately not reproved here.